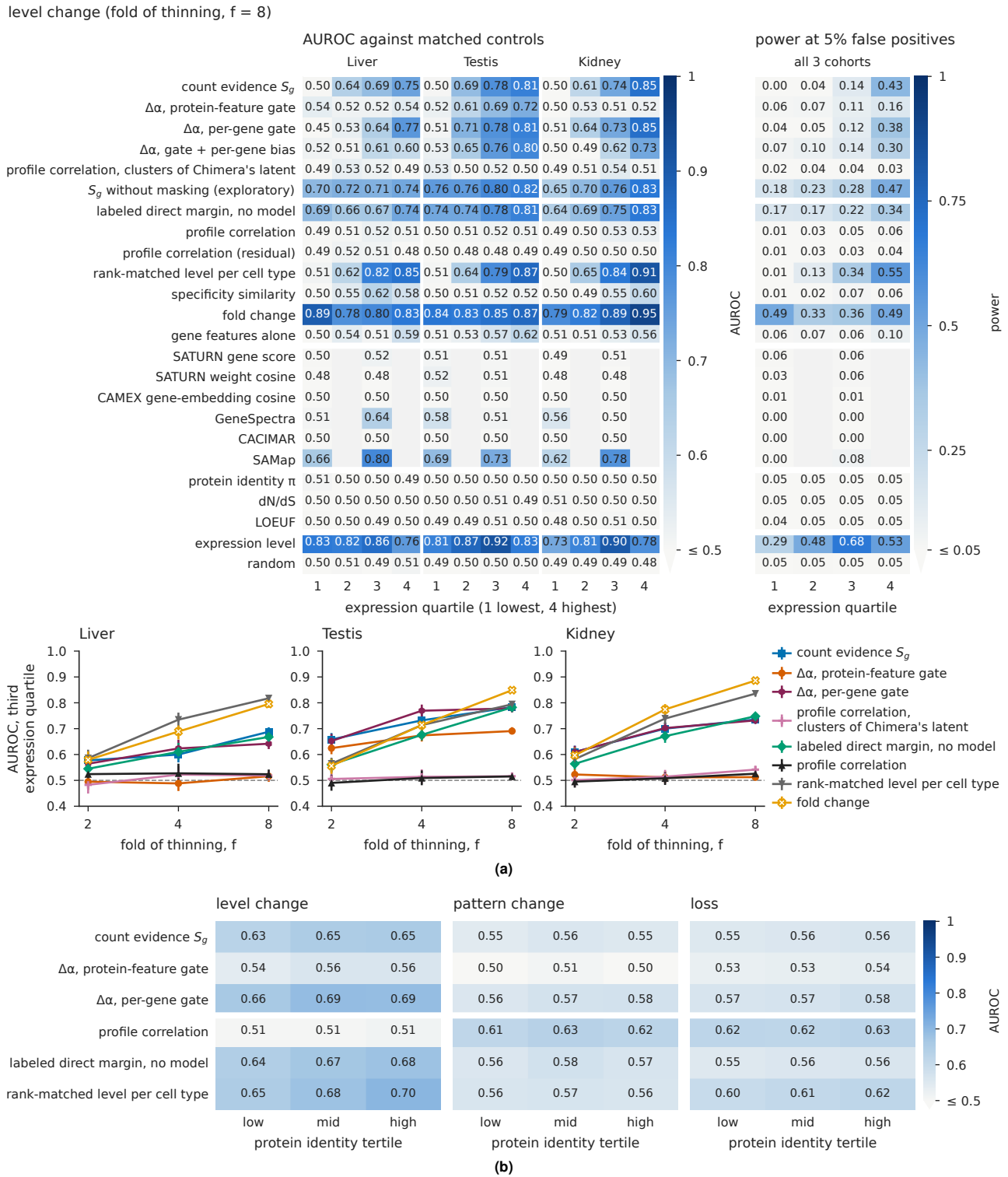

level change (fold of thinning, f = 8)
AUROC against matched controls
Liver
Testis
Kidney
power at 5% false positives
all 3 cohorts
count evidence S_g 0.50 0.64 0.69 0.75 0.50 0.69 0.78 0.81 0.50 0.61 0.74 0.85 0.00 0.04 0.14 0.43
Δα, protein-feature gate 0.54 0.52 0.52 0.54 0.52 0.61 0.69 0.72 0.50 0.53 0.51 0.52 0.06 0.07 0.11 0.16
Δα, per-gene gate 0.45 0.53 0.64 0.77 0.51 0.71 0.78 0.81 0.51 0.64 0.73 0.85 0.04 0.05 0.12 0.38
Δα, gate + per-gene bias 0.52 0.51 0.61 0.60 0.53 0.65 0.76 0.80 0.50 0.49 0.62 0.73 0.07 0.10 0.14 0.30
profile correlation, clusters of Chimera's latent 0.49 0.53 0.52 0.49 0.53 0.50 0.52 0.50 0.49 0.51 0.54 0.51 0.02 0.04 0.04 0.03
S_g without masking (exploratory) 0.70 0.72 0.71 0.74 0.76 0.76 0.80 0.82 0.65 0.70 0.76 0.83 0.18 0.23 0.28 0.47
labeled direct margin, no model 0.69 0.66 0.67 0.74 0.74 0.74 0.78 0.81 0.64 0.69 0.75 0.83 0.17 0.17 0.22 0.34
profile correlation 0.49 0.51 0.52 0.51 0.50 0.51 0.52 0.51 0.49 0.50 0.53 0.53 0.01 0.03 0.05 0.06
profile correlation (residual) 0.49 0.52 0.51 0.48 0.50 0.48 0.48 0.49 0.49 0.50 0.50 0.50 0.01 0.03 0.03 0.04
rank-matched level per cell type 0.51 0.62 0.82 0.85 0.51 0.64 0.79 0.87 0.50 0.65 0.84 0.91 0.01 0.13 0.34 0.55
specificity similarity 0.50 0.55 0.62 0.58 0.50 0.51 0.52 0.52 0.50 0.49 0.55 0.60 0.01 0.02 0.07 0.06
fold change 0.89 0.78 0.80 0.83 0.84 0.83 0.85 0.87 0.79 0.82 0.89 0.95 0.49 0.33 0.36 0.49
gene features alone 0.50 0.54 0.51 0.59 0.51 0.53 0.57 0.62 0.51 0.51 0.53 0.56 0.06 0.07 0.06 0.10
SATURN gene score 0.50 0.52 0.51 0.51 0.49 0.51 0.06 0.06
SATURN weight cosine 0.48 0.48 0.52 0.51 0.48 0.48 0.03 0.06
CAMEX gene-embedding cosine 0.50 0.50 0.50 0.50 0.50 0.50 0.01 0.01
GeneSpectra 0.51 0.64 0.58 0.51 0.56 0.50 0.00 0.00
CACIMAR 0.50 0.50 0.50 0.50 0.50 0.50 0.00 0.00
SAMap 0.66 0.80 0.69 0.73 0.62 0.78 0.00 0.08
protein identity π 0.51 0.50 0.50 0.49 0.50 0.50 0.50 0.50 0.50 0.50 0.50 0.50 0.05 0.05 0.05 0.05
dN/dS 0.50 0.50 0.50 0.50 0.50 0.50 0.51 0.49 0.51 0.50 0.50 0.50 0.05 0.05 0.05 0.05
LOEUF 0.50 0.50 0.49 0.50 0.49 0.49 0.51 0.50 0.48 0.50 0.51 0.50 0.04 0.05 0.05 0.05
expression level 0.83 0.82 0.86 0.76 0.81 0.87 0.92 0.83 0.73 0.81 0.90 0.78 0.29 0.48 0.68 0.53
random 0.50 0.51 0.49 0.51 0.49 0.50 0.50 0.50 0.49 0.49 0.49 0.48 0.05 0.05 0.05 0.05
1 2 3 4 1 2 3 4 1 2 3 4
expression quartile (1 lowest, 4 highest)
1 2 3 4
expression quartile
AUROC
power
1
0.9
0.8
0.7
0.6
≤ 0.5
0.75
0.5
0.25
≤ 0.05
Liver
Testis
Kidney
AUROC, third expression quartile
1.0
0.9
0.8
0.7
0.6
0.5
0.4
2
4
8
fold of thinning, f
count evidence S_g
Δα, protein-feature gate
Δα, per-gene gate
profile correlation, clusters of Chimera's latent
labeled direct margin, no model
profile correlation
rank-matched level per cell type
fold change
(a)
level change
pattern change
loss
count evidence S_g 0.63 0.65 0.65 0.55 0.56 0.55 0.55 0.56 0.56
Δα, protein-feature gate 0.54 0.56 0.56 0.50 0.51 0.50 0.53 0.53 0.54
Δα, per-gene gate 0.66 0.69 0.69 0.56 0.57 0.58 0.57 0.57 0.58
profile correlation 0.51 0.51 0.51 0.61 0.63 0.62 0.62 0.62 0.63
labeled direct margin, no model 0.64 0.67 0.68 0.56 0.58 0.57 0.55 0.56 0.56
rank-matched level per cell type 0.65 0.68 0.70 0.56 0.57 0.56 0.60 0.61 0.62
low mid high
protein identity tertile
AUROC
1
0.9
0.8
0.7
0.6
≤ 0.5
(b)